

DAY-6

TASK:3 Research on Clothing Segmentation AI

Today, I researched Clothing Segmentation AI and understood how AI systems separate clothing items from the human body and background in images and videos. I learned that clothing segmentation is an important computer vision technique used in virtual try-on systems and fashion AI applications.

How I Understood It Works

1. Image Input

I learned that the AI system first takes an image or video frame containing a person.

2. Feature Extraction

The deep learning model then analyzes:

- Colors
- Edges
- Textures
- Clothing boundaries
- Body structure

3. Semantic Segmentation

I found that the image gets divided into different labeled regions such as:

- Shirt
- Pants
- Face
- Arms
- Background

4. Pixel Classification

The AI assigns every pixel to a category.

Example:

Blue pixels → jeans

Skin-tone pixels → face/arms

White pixels → shirt

5. Output Mask Generation

Finally, the model creates segmentation masks which are later used in virtual try-on and fashion applications.

There are different types of segmentation that I came across:

1. Semantic Segmentation

I understood that this detects clothing categories but does not separate individual objects.

2. Instance Segmentation

This separates multiple clothing items individually.

3. Human Parsing

I found this especially important because it provides detailed labeling of body parts and garments.

Applications of Cloth segmentation:

1. Virtual Try-On
2. Size Recommendation
3. Background Removal
4. Smart Mirrors

CHALLENGES FACED:

While researching, I observed that clothing segmentation models still face several challenges in real-world applications. Loose or oversized clothing can make it difficult for the AI to accurately identify body boundaries, while backgrounds with colors similar to the clothing may confuse the segmentation process. I also found that occluded body parts, complex textures, patterned garments, and low lighting conditions can reduce segmentation accuracy and affect the quality of the final output. At the same time, I understood that clothing segmentation is one of the most important technologies used in virtual try-on systems because it helps detect garment boundaries accurately, preserves the body structure of the user, improves the realism of virtual fitting, and reduces clothing overlap errors. Without proper segmentation, virtual try-on results can appear unrealistic and inaccurate, which negatively affects the overall user experience.